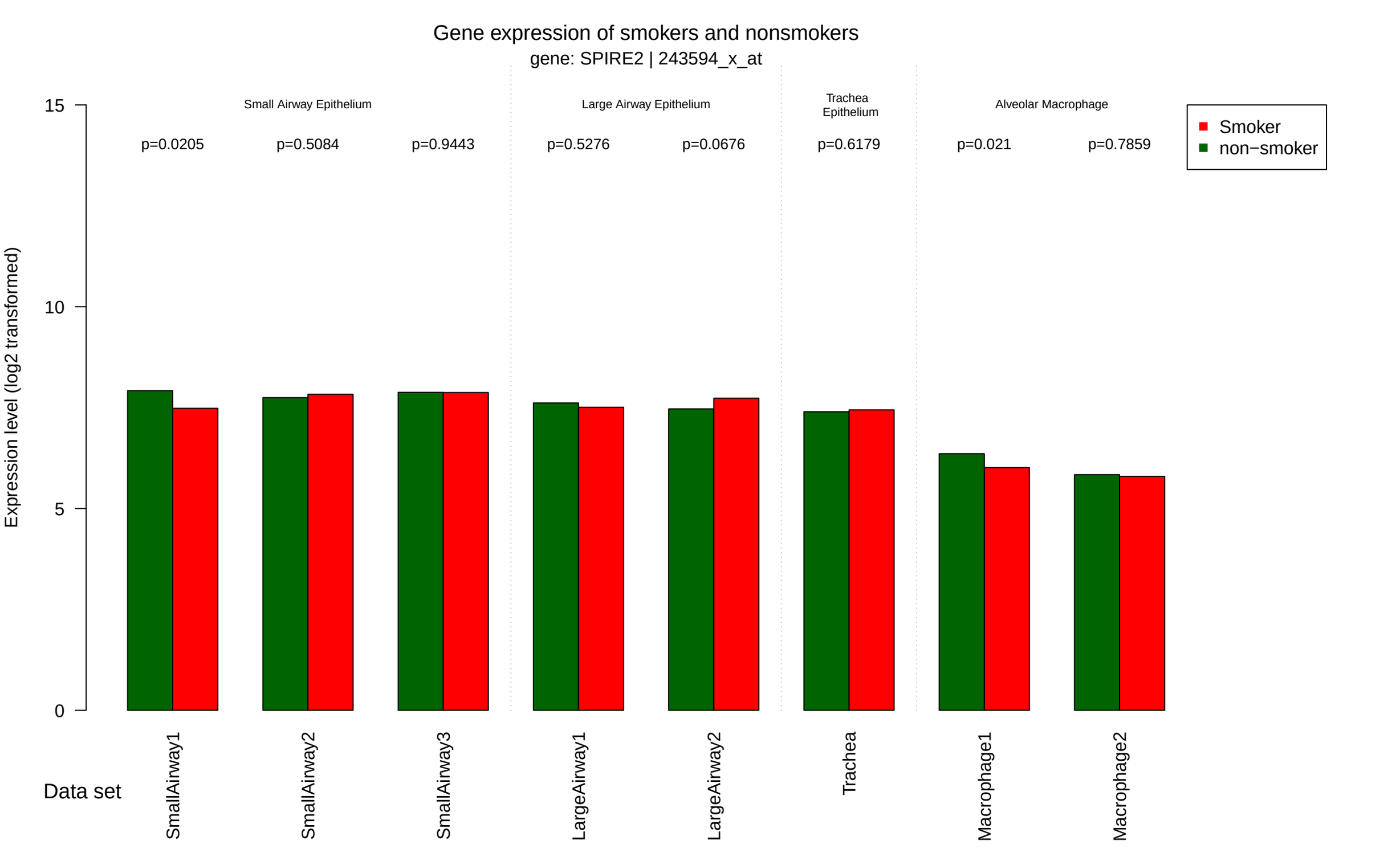
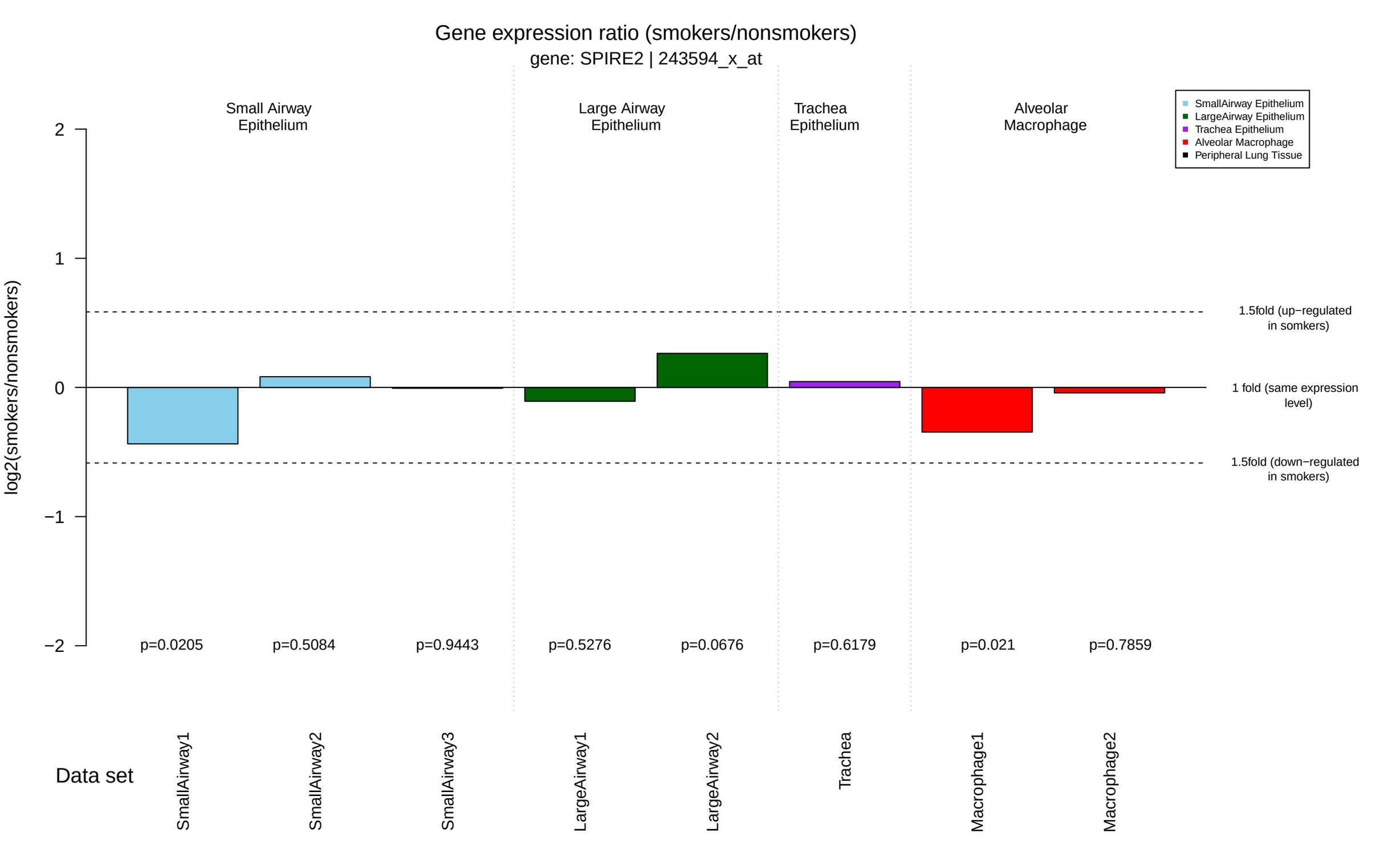
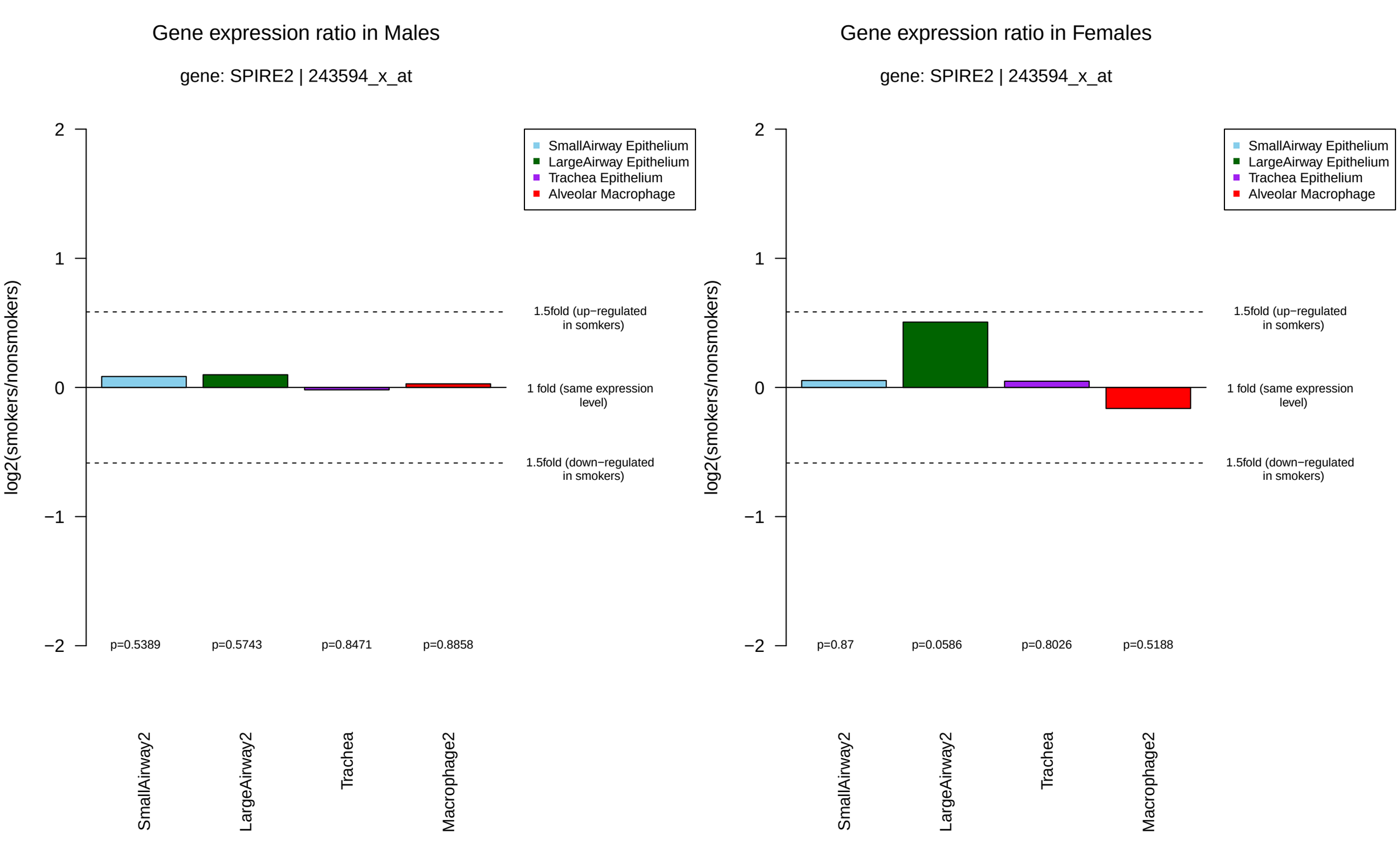
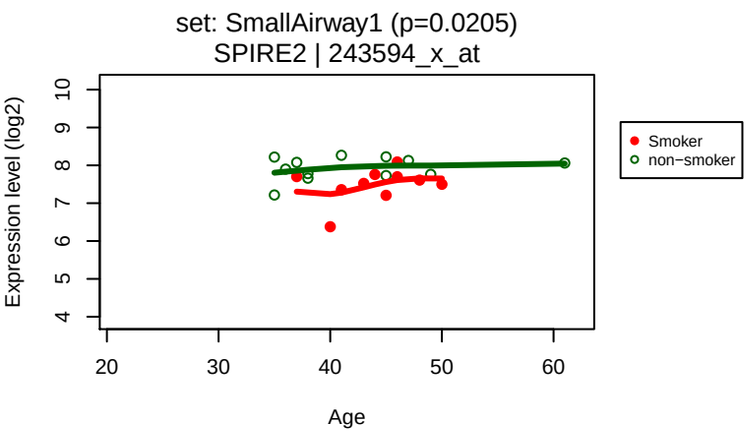
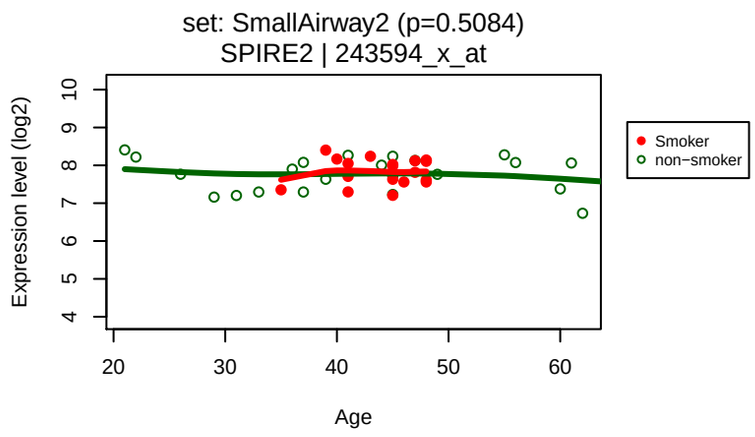
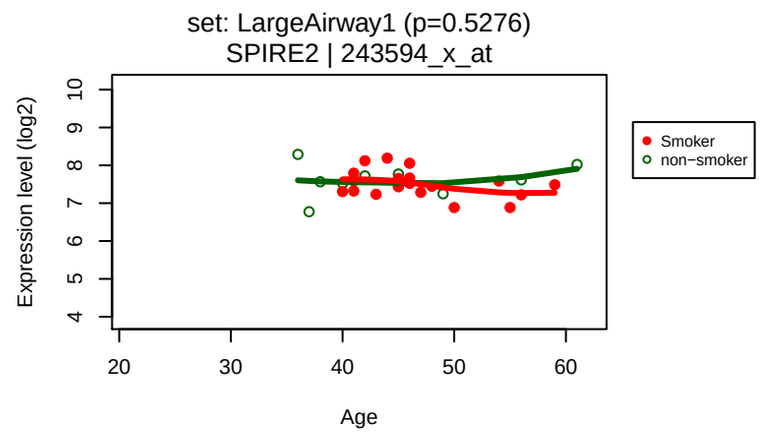

Gene expression of smokers and nonsmokers
gene: SPIRE2 | 243594_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0205
p=0.5084
p=0.9443
p=0.5276
p=0.0676
p=0.6179
p=0.021
p=0.7859
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

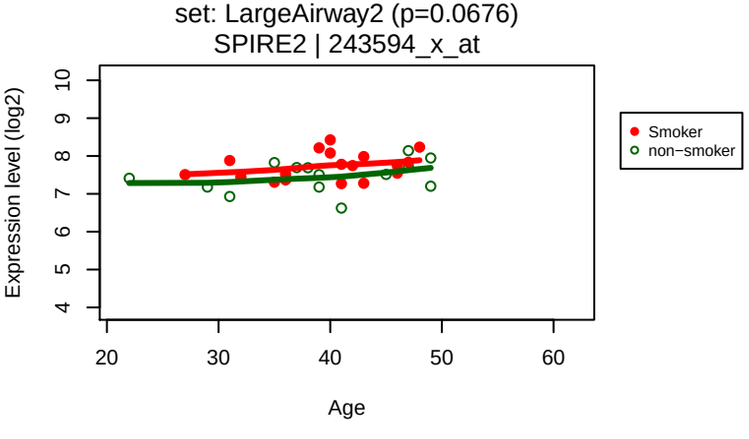

Gene expression ratio (smokers/nonsmokers)
gene: SPIRE2 | 243594_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0205
p=0.5084
p=0.9443
p=0.5276
p=0.0676
p=0.6179
p=0.021
p=0.7859
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

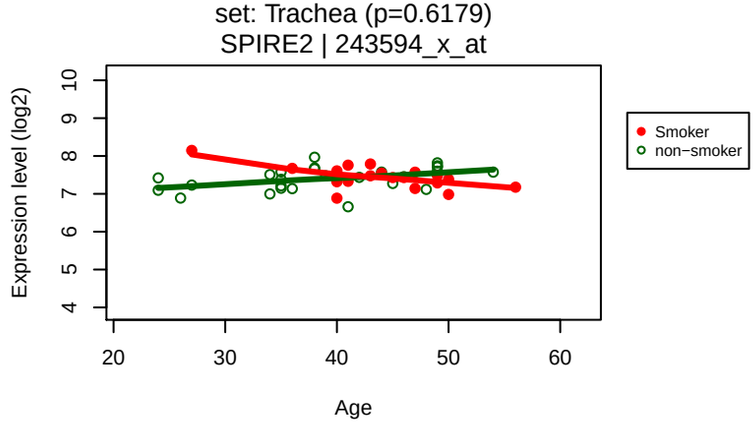

Gene expression ratio in Males
gene: SPIRE2 | 243594_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5389
p=0.5743
p=0.8471
p=0.8858
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPIRE2 | 243594_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.87
p=0.0586
p=0.8026
p=0.5188
SmallAirway2
LargeAirway2
Trachea
Macrophage2

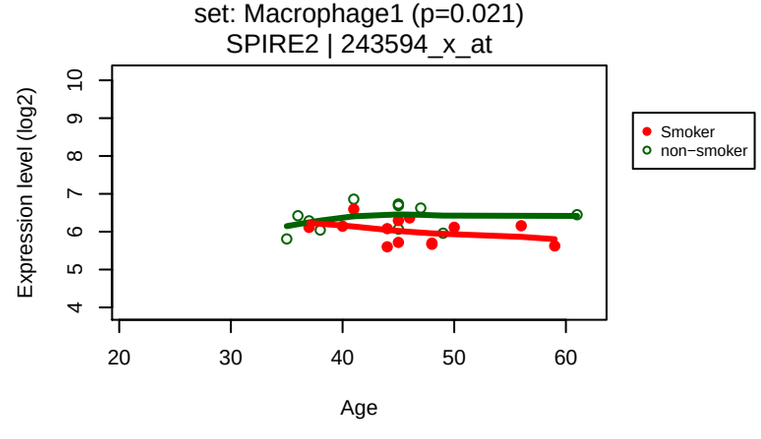

set: SmallAirway1 (p=0.0205)
SPIRE2 | 243594_x_at
Expression level (log2)
Age
Smoker
non−smoker

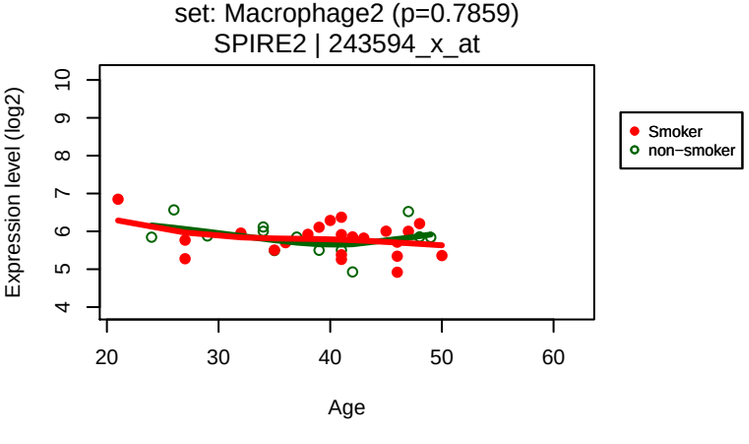

set: SmallAirway2 (p=0.5084)
SPIRE2 | 243594_x_at
Expression level (log2)
Age
Smoker
non−smoker

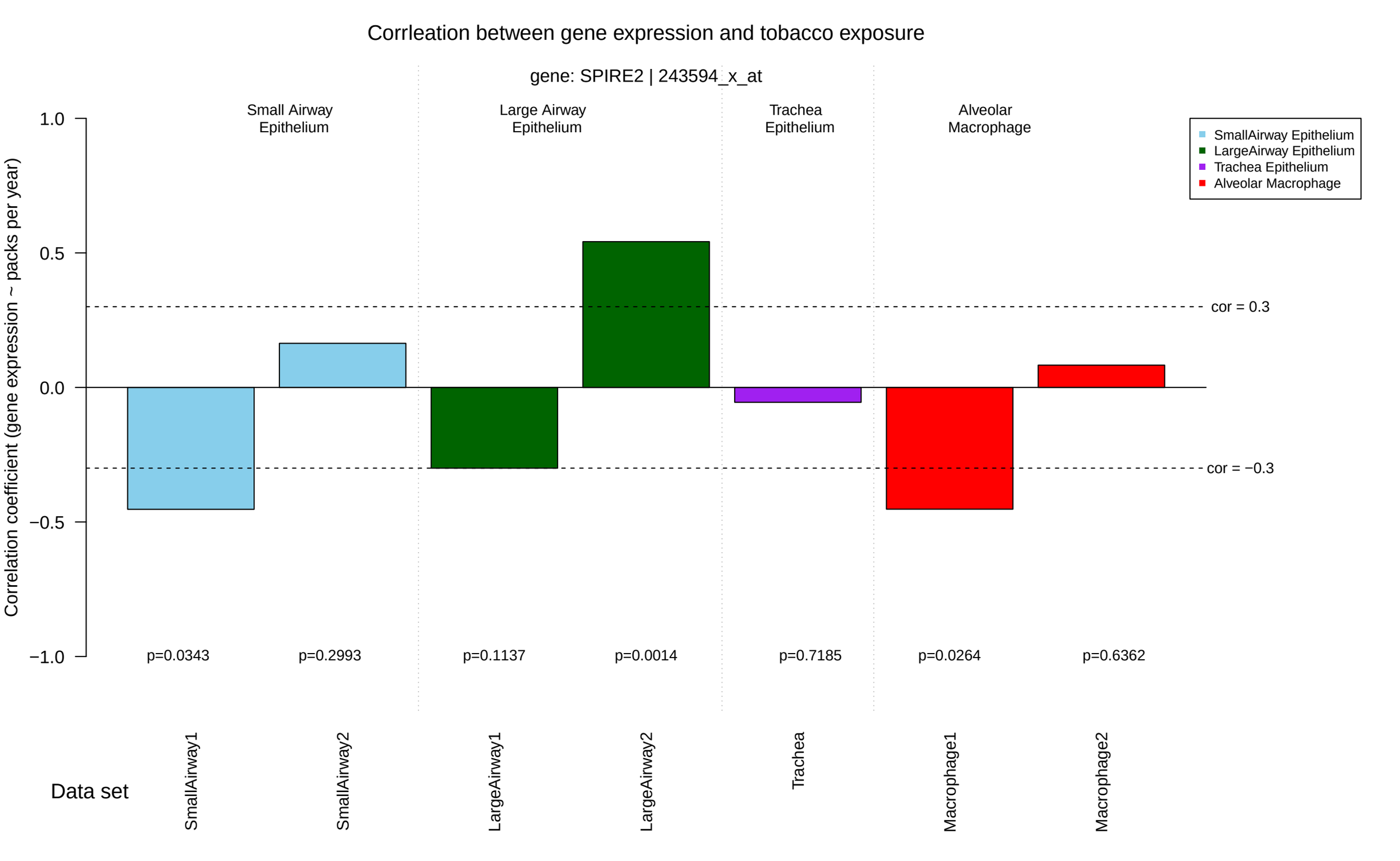

set: LargeAirway1 (p=0.5276)
SPIRE2 | 243594_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0676)
SPIRE2 | 243594_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6179)
SPIRE2 | 243594_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.021)
SPIRE2 | 243594_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7859)
SPIRE2 | 243594_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPIRE2 | 243594_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0343
p=0.2993
p=0.1137
p=0.0014
p=0.7185
p=0.0264
p=0.6362
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2